Experiment Title: Practice of Basic Linux Terminal Commands

Topic: File and Directory Navigation in Linux Terminal

1. Objective

To practice basic Linux navigation and file listing commands:

- Check current user (`whoami`) and location (`pwd`).
- Move between directories (`cd`, `cd ~`).
- List files, hidden files, and details (`ls`, `-l`, `-a`, `-lh`, `-la`).

2. Theoretical Background

Linux uses a single directory tree starting at root (`/`). Users start in home (`~`). Commands navigate and inspect files.

2.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`whoami`	`whoami`	Print current username.	`whoami`
`pwd`	`pwd`	Print current directory path.	`pwd`
`cd`	`cd [dir]`	Move to directory.	`cd lab_demo`
`cd ~`	`cd ~`	Go to home directory.	`cd ~`
`ls`	`ls`	List visible files and folders.	`ls`
`ls -l`	`ls -l`	Long listing with file details.	`ls -l`
`ls -a`	`ls -a`	List all files, including hidden (`.` prefix).	`ls -a`
`ls -lh`	`ls -lh`	Show human-readable sizes (K, M, G).	`ls -lh`
`ls -la`	`ls -la`	Show all files with full details.	`ls -la`

2.2 Understanding File Details (`ls -l`)

Running `ls -l` shows six key fields for each file:

Field	Simple Explanation
1. File Type & Permissions	Type (`-` file, `d` folder) and access rights (`rwx`) for owner, group, others.
2. Links	Number of hard links.
3. Owner & Group	User and group that own the file.
4. File Size	Size in bytes (or KB/MB with `-h`).
5. Date & Time	Last modification timestamp.
6. File Name	Name of file or folder.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul`	**Command Shell**	GNU Bash 5.2 / Zsh
Terminal Emulator	Linux Terminal (Kitty / Ghostty)	**Core Tools**	GNU Coreutils (`ls`, `pwd`, `whoami`)

4. Procedure / Steps

Steps followed in this experiment:

1. Check current user (**whoami**) and directory path (**pwd**).
2. List home directory contents (**ls**).
3. Move into demo folder (**cd lab_demo**) and verify with **pwd**.
4. List normal files (**ls**) and hidden files (**ls -a**).
5. Inspect file details in bytes (**ls -l**) and readable units (**ls -lh**).
6. List all files with full attributes (**ls -la**).
7. Return to home directory (**cd ~**).

5. Result / Output Demonstration

All commands were executed in the terminal under user `ratul` on machine `firefly`.

5.1 Session 1: Finding Our Location, Moving Folders, and Listing Files

ratul@firefly: ~ — bash

```
ratul@firefly:~$ whoami
ratul
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ ls
Desktop  Documents  Downloads  lab_demo  Music  Pictures  Videos
ratul@firefly:~$ cd lab_demo
ratul@firefly:~/lab_demo$ pwd
/home/ratul/lab_demo
ratul@firefly:~/lab_demo$ ls
backups  data.csv  notes.txt  report.pdf  scripts
ratul@firefly:~/lab_demo$ ls -a
.  ..  .lab_profile  .secret.key  backups  data.csv  notes.txt  report.pdf  scripts
ratul@firefly:~/lab_demo$ _
```

Explanation of Session 1 Results:

- **whoami** and **pwd**: Identified user `ratul` at `/home/ratul`.
- **cd lab_demo**: Moved into folder; prompt updated to `~/lab_demo`.
- **ls** vs **ls -a**: **ls** shows visible files; **-a** reveals hidden files (`.lab_profile`, `.secret.key`) and directory links (`.`, `..`).

5.2 Session 2: Inspecting File Details, Easy Sizes, and Returning Home

ratul@firefly: ~/lab_demo — bash

```
ratul@firefly:~/lab_demo$ ls -l
total 23568
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r-- 1 ratul users 14680064 Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ ls -lh
total 24M
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:15 backups
-rw-r--r-- 1 ratul users  14M Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users 1.4K Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users 256K Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users 4.0K Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ ls -la
total 23580
drwxr-xr-x 4 ratul users     4096 Oct  8 10:28 .
drwx------ 9 ratul users     4096 Oct  8 09:50 ..
-rw-r--r-- 1 ratul users      342 Oct  8 10:12 .lab_profile
-rw------- 1 ratul users       64 Oct  8 10:14 .secret.key
drwxr-xr-x 2 ratul users     4096 Oct  8 10:15 backups
-rw-r--r-- 1 ratul users 14680064 Oct  8 10:20 data.csv
-rw-r--r-- 1 ratul users     1433 Oct  8 10:22 notes.txt
-rw-r--r-- 1 ratul users   262144 Oct  8 10:25 report.pdf
drwxr-xr-x 2 ratul users     4096 Oct  8 10:18 scripts

ratul@firefly:~/lab_demo$ cd ~
ratul@firefly:~$ pwd
/home/ratul
ratul@firefly:~$ _
```

Explanation of Session 2 Results:

- `ls -l` vs `ls -lh`: `-l` shows raw bytes; `-lh` converts to human-readable units (`14M`).
- `ls -la`: Shows full details, including permissions on hidden files (`.secret.key`).
- `cd ~`: Returned to home directory (`/home/ratul`).

6. Conclusion

Key takeaways from this lab:

1. Navigated directories using **`pwd`**, **`cd`**, and **`cd ~`**.
2. Generated standard, hidden, and detailed file listings using **`ls`** options.
3. Identified file permissions, ownership, sizes, and timestamps.